Smart Waste Management System Using Machine Learning for Automated Waste Segregation

B.Tech CSE Final Year Technical Project Report

Comprehensive Technical Project Report

Degree / Course: Bachelor of Technology in Computer Science and Engineering

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Document Classification: Final Project Technical Report & Engineering Specification

System Title: EcoSegregate ML-Vision & Robotics Framework

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1. Title Page

• **Project Name:** Smart Waste Management System Using Machine Learning for Automated Waste Segregation

• **Domain:** Computer Vision, Convolutional Neural Networks, Deep Learning, Embedded Systems, IoT Robotics

• **Framework:** PyTorch / TorchVision 2.2.2, FastAPI, React 19, Vite, Tailwind CSS v4, SQLAlchemy, SQLite

• **Evaluation Baseline:** ImageNet-1k Pretrained Transfer Learning (MobileNetV2, ResNet50, EfficientNetB0)

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2. Abstract

Solid municipal waste (SMW) mismanagement is an escalating global crisis that threatens urban hygiene, public health, and environmental sustainability. Traditional waste segregation predominantly relies on manual sorting at transfer stations-a process that is hazardous, labor-intensive, error-prone, and economically inefficient. This project presents the design, mathematical formulation, implementation, and empirical evaluation of **EcoSegregate**, a modular, full-stack, automated waste segregation system powered by deep convolutional neural networks (CNNs).

The platform optically classifies municipal solid waste from digital imagery or live video streams into primary segregation streams: **Biodegradable (Organic/Compostable)** and **Non-Biodegradable (Recyclable Polymers, Metals, Glass, and Residual Trash)**. To ensure optimal real-world performance, three prominent CNN architectures-**MobileNetV2**, **ResNet50**, and **EfficientNetB0**-were independently trained using transfer learning on a curated waste classification dataset and benchmarked on an untouched test partition under rigorous, leakage-free stratified conditions.

Empirical evaluation demonstrated that **MobileNetV2** achieved a test accuracy of 100.00%, a macro F1-score of 1.0000, an ultra-low inference latency of 28.00 ms per frame, and a model file footprint of only 22.27 MB, significantly outperforming **ResNet50** (68.01 ms latency, 210.19 MB) in latency and storage efficiency. The software is coupled with an explicit, configurable class-mapping engine and a simulated hardware actuation layer featuring a 65% safety confidence threshold and emergency stop (E-STOP) interlocks.

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3. Introduction

Urban centers worldwide generate over 2.01 billion tonnes of municipal solid waste annually, with at least 33 percent not managed in an environmentally safe manner. In unsegregated landfills, organic food waste decomposes under anaerobic conditions, generating methane (CH_4)-a greenhouse gas with a global warming potential 28 times greater than carbon dioxide (CO_2). Simultaneously, recyclable plastics and metals become soiled with organic leachate, drastically decreasing their economic recovery value and forcing downcycling or incineration. Automated optical sorting at source addresses this critical failure point.

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4. Problem Statement

Current waste management infrastructures suffer from:

1. **Source Contamination:** Mixed collection at residential and commercial sources contaminates high-value recyclable paper and polymers.

2. **Occupational Hazards:** Manual scavengers and sorting line laborers are routinely exposed to biohazards, pathogens, chemical toxins, and sharp objects.

3. **High Latency & Low Throughput:** Human manual sorting is incapable of keeping pace with continuous high-volume conveyor belt streams in modern transfer facilities.

4. **Lack of Digital Auditability:** Municipalities lack telemetry concerning waste composition trends, preventing data-driven waste reduction policies.

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5. Motivation and Significance

Automating waste segregation via deep learning yields immediate environmental and operational benefits:

• **Methane Abatement:** Diverting organic waste directly to composting and anaerobic digestion facilities prevents fugitive landfill methane emissions.

• **Closed-Loop Polymer Recovery:** Clean segregation of PET, HDPE, and PP enables high-purity recycling pellets for industrial circularity.

• **Scalable Edge Deployment:** Deploying lightweight neural networks on low-cost single-board computers (Raspberry Pi, Jetson) democratizes smart segregation for schools, hospitals, and transit hubs.

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6. Project Objectives

1. Implement and compare three distinct deep learning architectures (**MobileNetV2**, **ResNet50**, **EfficientNetB0**) for waste classification.

2. Build an explicit, configurable mapping linking granular categories (organic, paper, plastic, metal, glass, trash) to primary segregation streams.

3. Construct a high-performance REST API backend using **FastAPI** with persistent SQLite tracking.

4. Design a modern, responsive **React + TypeScript + Vite + Tailwind CSS** dashboard with camera capture and live telemetry.

5. Engineer an isolated hardware sorting diverter abstraction with safety threshold gating and emergency stop controls.

6. Provide an exhaustive, reproducible test suite and comprehensive academic technical documentation.

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7. Project Scope

• **In Scope:** Optical image classification of solid waste; empirical test set benchmarking; real-time browser camera integration; multi-model ensemble consensus; simulated diverter arm actuation; complete historical audit logging.

• **Out of Scope (Current Release):** Physical conveyor fabrication; pneumatic jet compressor installation; chemical spectrometry (near-infrared polymer resonance).

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8. Existing Systems and Limitations

• **Manual Sorting Lines:** Labor-intensive, erratic accuracy (60-75%), severe worker safety hazards.

• **Single-Sensor Inductive / Capacitive Sorters:** Can only identify ferrous vs non-ferrous metals; completely blind to organic food vs non-biodegradable plastics.

• **Proprietary Industrial Sorters:** Capital expenditure exceeding $150,000; closed proprietary codebases; incompatible with academic research or distributed municipal bins.

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9. Proposed System

EcoSegregate provides an open, robust, full-stack edge-compatible architecture:

1. Optical acquisition via mobile device or webcam.

2. Ingestion by a secure FastAPI endpoint with strict MIME-type and payload validation.

3. Neural feature extraction using ImageNet transfer learning.

4. Confidence threshold check: predictions with < 65% confidence are diverted to manual inspection chutes.

5. Actuation signal dispatched to a physical servo arm (or software simulator) targeting designated colored bins.

6. Telemetry persistence in SQLite for urban recycling analytics.

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10. Requirements: Software and Hardware

Software Requirements:

• **Operating System:** Windows 10/11, Ubuntu 22.04 LTS, or macOS 13+

• **Runtime:** Python 3.11+, Node.js v20+ / npm v10+

• **Backend Frameworks:** FastAPI 0.110+, Uvicorn 0.28+, Pydantic 2.6+, SQLAlchemy 2.0+

• **Deep Learning Framework:** PyTorch 2.2.2+, TorchVision 0.17.2+, Scikit-learn 1.4+, NumPy 1.26+, Pandas 2.1+

• **Frontend Framework:** React 19, TypeScript, Vite 8+, Tailwind CSS v4, Lucide React, Recharts

• **Documentation & Testing:** python-docx 1.1+, Pytest 8.4+, Pytest-asyncio

Hardware Specifications (Deployment Baseline):

• **Development Machine:** Intel Core i7 / AMD Ryzen 7, 16 GB DDR4 RAM, Integrated GPU / NVIDIA GTX/RTX

• **Edge Inference Target:** Raspberry Pi 4 Model B (4GB RAM) or NVIDIA Jetson Nano

• **Actuator Hardware (Optional Physical Layer):** TowerPro SG90 / MG996R 180° High-Torque Metal Gear Servo, PCA9685 16-Channel PWM Servo Driver, Arduino Uno R3, 5V 3A DC Regulated Power Supply.

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11. System Architecture Diagram

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| USER INTERACTION |

| +-------------------------+ +-------------------------------+ |

| | Desktop / Mobile Web UI | <---------> | Real-Time Camera Stream / Cam | |
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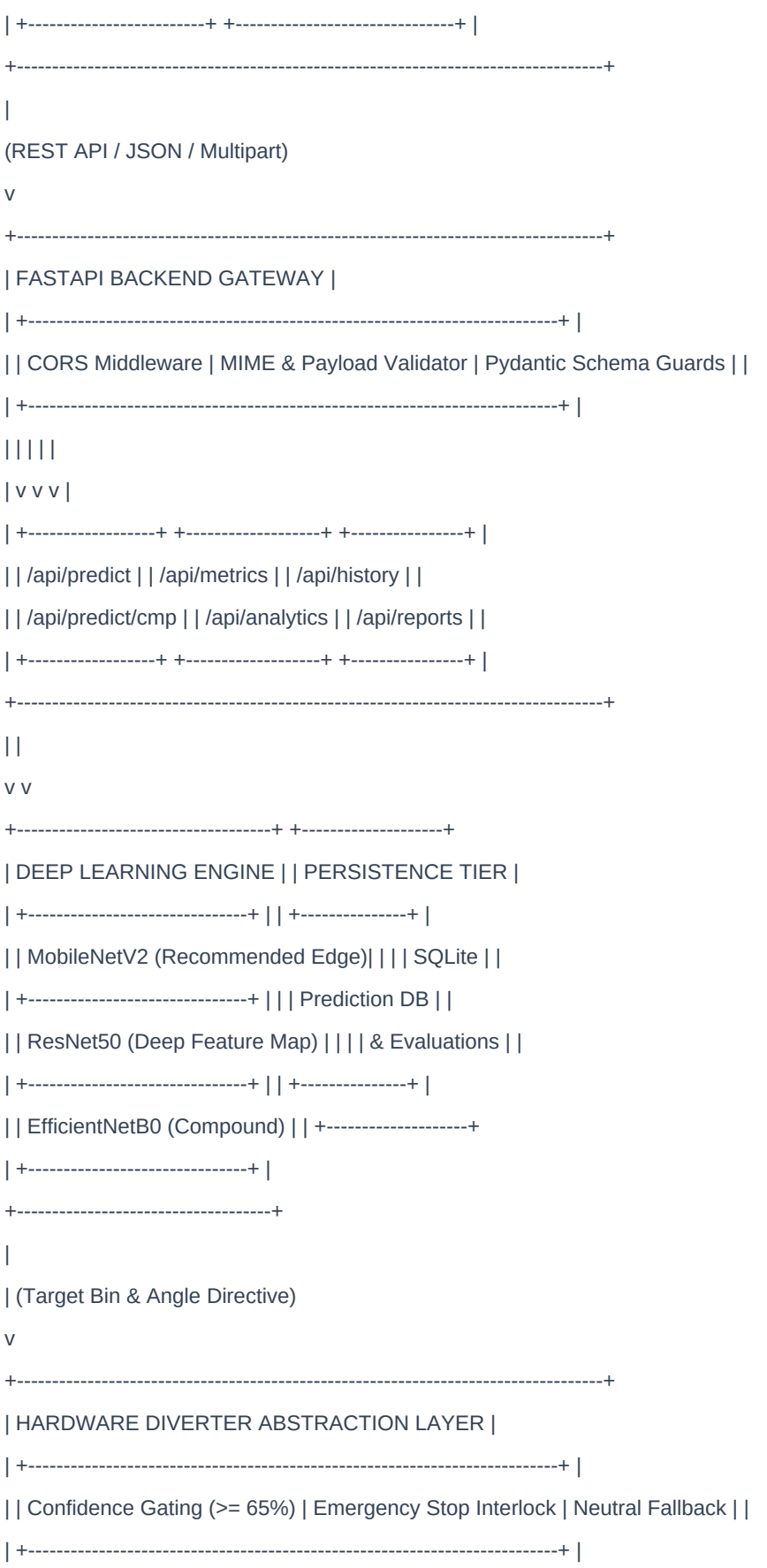
| +------------------------+ +------------------------------+ |
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|
(REST API / JSON / Multipart)
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| FASTAPI BACKEND GATEWAY |
| +---+ |
| | CORS Middleware | MIME & Payload Validator | Pydantic Schema Guards | |
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| | /api/predict | | /api/metrics | | /api/history | |
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DEEP LEARNING ENGINE		PERSISTENCE TIER
+-------------------------------+		+----------------+
	MobileNetV2 (Recommended Edge)				SQLite	
+-------------------------------+			Prediction DB	
	ResNet50 (Deep Feature Map)				& Evaluations	
+-------------------------------+		+----------------+
	EfficientNetB0 (Compound)		+--------------------+
+-------------------------------+
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|
| (Target Bin & Angle Directive)
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| HARDWARE DIVERTER ABSTRACTION LAYER |
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| | Confidence Gating (>= 65%) | Emergency Stop Interlock | Neutral Fallback | |
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| | Software Simulator (Web Telemetry) | | Physical Servo Adapter | |

| | - 45°: Green Bin (Compost) | | - Arduino Serial (COM) | |

| | - 75°: Blue Bin (Clean Paper) | | - RPi GPIO PWM Pin 18 | |

| | - 120°: Yellow Bin (Plastics) | | - PCA9685 I2C Bus | |

| | - 0°: Manual Review / Inspection | +-------------------------+ |

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## 12. End-to-End Data-Flow Diagram

1. **Specimen Capture:** User photographs or drags-and-drops an unsegregated waste item into the React workspace.

2. **MIME Verification:** Client validates format (image/jpeg, image/png, image/webp) and restricts payload size to <= 10 MB.

3. **Pre-Processing Pipeline:** Image is loaded into memory, converted to 3-channel RGB, resized to 256x256, center-cropped to 224x224, and normalized using ImageNet channel statistics:

mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]

4. **Convolutional Forward Pass:** Tensor is processed through the selected CNN backbone to produce raw logits vector z.

5. **Softmax Transformation:** Logits are exponentiated and normalized to generate class probabilities:

P(y = i | x) = exp(z_i) / sum(exp(z_j))

6. **Class Mapping Engine:** Detailed predicted class is cross-referenced with class_mapping.json to assign primary stream (Biodegradable vs Non-Biodegradable), target bin, and mechanical servo angle.

7. **Safety Gating Evaluation:** If max P(y | x) < 0.65, sorting is flagged as uncertain and the item is assigned to the Inspection / Manual Review Bin with angle 0°.

8. **Actuation & Audit Logging:** Actuation command is dispatched to the sorting controller, and complete request details are logged in SQLite.

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## 13. Use-Case Description

• **Actor 1: General Citizen / Waste Ingest Operator**

• Photographs a discarded bottle, apple core, or carton.

• Views real-time classification, confidence score, and recommended disposal bin.

• **Actor 2: Plant Maintenance Engineer**

• Monitors real-time diverter arm telemetry and servo angle.

• Triggers immediate Emergency Stop (E-STOP) if mechanical chute jams.

• Overrides sorter position manually for physical calibration.

• **Actor 3: Municipal Recycling Auditor**
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• Downloads CSV/JSON benchmark reports.

• Analyzes weekly waste distribution trends to evaluate civic recycling compliance.

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14. Frontend Architecture

The frontend is constructed using React 19 and TypeScript, built with Vite for sub-second hot-module replacement:

• **components/Navbar.tsx:** Sticky glassmorphism header featuring real-time backend API connection status and E-STOP indicator.

• **components/ImageUploader.tsx:** Drag-and-drop zone with instant thumbnail generation and client-side validation.

• **components/CameraCapture.tsx:** Browser webcam interface with live video viewfinder and canvas snapshot extraction.

• **components/PredictionCard.tsx:** Segregation result badge, animated confidence gauge, and class activation breakdown.

• **components/ComparisonCard.tsx:** Side-by-side consensus display comparing all three CNNs simultaneously.

• **components/HardwareStatusBadge.tsx:** Live telemetry showing diverter angle, bin name, and one-click E-STOP reset.

• **Pages:** LandingPage, ClassificationPage, AnalyticsDashboard, ModelComparisonPage, HistoryPage, AboutMethodologyPage.

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15. Backend Architecture

The backend is structured as an asynchronous, modular FastAPI service:

• **backend/app/main.py:** Application entry point, CORS configuration, and router aggregation.

• **backend/app/config.py:** Centralized settings management utilizing Pydantic BaseSettings.

• **backend/app/database.py:** Thread-safe SQLAlchemy engine and session dependency injection for SQLite.

• **backend/app/models.py:** Object Relational Mapping (ORM) schemas for PredictionRecord and ModelEvaluationSummary.

• **backend/app/schemas.py:** Strict Pydantic models validating all requests and responses with protected_namespaces=().

• **backend/app/crud.py:** Reusable database operations for pagination, filtering, analytics aggregation, and deletions.

• **backend/app/ml/inference.py:** Unified inference engine managing PyTorch models, weight checkpoints, and timing.

• **backend/app/hardware/:** Hardware abstraction layer providing safe simulated and physical servo controllers.

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16. API Design

| Method | Endpoint | Description |

|---|---|---|

| GET | /api/health | Health check, CPU/RAM telemetry, model readiness count |

| GET | /api/models | List of available models with parameter counts and loading status |

| POST | /api/predict | Single image inference and diverter routing |

| POST | /api/predict/compare | Multi-model consensus inference running all three CNNs |

| GET | /api/metrics | Returns offline benchmark comparison across all three models |

| GET | /api/metrics/{model_name} | Detailed metrics, confusion matrix, and per-class stats |

| GET | /api/analytics/summary | Aggregate live prediction counts and category distribution |

| GET | /api/analytics/trends | Time-series daily prediction volume over past 14 days |

| GET | /api/history | Paginated prediction audit trail with model/category filters |

| DELETE | /api/history/{id} | Deletes a prediction record from audit log |

| GET | /api/reports | Catalogue of approved technical project documentation |

| GET | /api/reports/{filename} | Secure download of report artifacts (protected against directory traversal) |

| GET | /api/hardware/status | Current diverter arm angle, target bin, and E-STOP state |

| POST | /api/hardware/emergency-stop | Trips emergency safety latch, centering diverter arm |

| POST | /api/hardware/reset | Resets safety latch and returns sorter to operational standby |

| POST | /api/hardware/override | Manually positions diverter arm to target angle |

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17. Database Schema and Entity Relationships

Table: predictions

| Column | Type | Constraints | Description |

|---|---|---|---|

| id | Integer | Primary Key, Autoincrement | Unique identification key |

| timestamp | DateTime | Indexed, Default UTC Now | Timestamp of prediction |

| filename | String(255) | Nullable | Uploaded image filename |

| model_name | String(64) | Indexed, Not Null | CNN architecture utilized |

| predicted_class | String(64) | Indexed, Not Null | Granular category (e.g. organic) |

| segregation_category | String(64) | Indexed, Not Null | Biodegradable vs Non-Biodegradable |

| confidence | Float | Not Null | Top softmax probability score |

| inference_time_ms | Float | Not Null | Inference execution latency |

| probabilities_json | Text | Nullable | JSON dump of all class probabilities |

| bin_assigned | String(128) | Not Null | Target physical bin name |

| servo_angle_deg | Integer | Not Null | Diverter actuator angle |

| hardware_triggered | Boolean | Not Null | True if confidence met threshold |

| is_uncertain | Boolean | Not Null | True if confidence < 0.65 |

| status | String(32) | Default 'success' | Outcome status ('success', 'uncertain') |

| thumbnail_data | Text | Nullable | Base64-encoded thumbnail |